

MOLOKAI, HI, USA

WMO: 911860

Lat: 21.155N

Lon: 157.096W

Elev: 443

StdP: 14.46

Time Zone: -10.00 (W10)

Period: 94-19

WBAN: 22534

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	59.3	61.4	49.5	53.4	70.4	52.6	59.9	68.2	25.6	73.8	24.1	73.9	2.1	50	0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.7	87.6	74.5	86.1	73.9	84.9	73.3	76.6	83.2	75.8	82.4	75.1	81.8	12.9	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.8	132.6	79.3	73.6	127.2	79.0	72.9	123.9	78.7	40.3	83.5	39.4	82.3	38.7	81.9	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.0	21.7	20.2	55.3	90.0	1.8	3.0	54.0	92.1	52.9	93.9	51.9	95.5	50.5	97.7		
(5)			52.7	77.7	2.3	2.7	51.0	79.6	49.7	81.2	48.4	82.8	46.7	84.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	75.0	71.1	71.0	71.6	73.1	74.8	77.1	78.4	79.0	78.7	77.5	75.3	72.8
	DBStd	3.70	2.81	2.55	2.69	2.29	2.12	1.60	1.96	1.73	2.00	2.13	2.28	2.65
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	1	0	0	1	0	0	0	0	0	0	0	0	0
	CDD50	9143	653	588	670	693	769	812	879	900	860	854	758	707
	CDD65	3669	188	169	205	243	304	362	414	435	410	389	308	242
	CDH74	20749	592	489	664	907	1613	2247	2910	3281	3112	2593	1478	863
	CDH80	3750	30	18	40	52	190	412	664	816	768	528	175	57
(14) Wind	WSAvg	10.9	9.3	9.8	10.4	11.7	10.5	12.3	13.0	12.6	10.7	10.3	10.4	9.8
(15) Precipitation	PrecAvg	29.1	4.9	3.9	3.0	2.4	1.7	0.6	0.7	0.6	0.7	2.2	3.3	4.7
	PrecMax	44.0	12.0	14.3	14.4	7.2	7.1	1.9	1.5	1.7	2.2	7.4	11.9	11.8
	PrecMin	12.9	0.6	0.3	0.2	0.4	0.0	0.0	0.0	0.0	0.1	0.0	0.2	0.0
	PrecStd	10.0	3.7	3.1	3.0	2.0	1.8	0.6	0.4	0.5	0.5	2.2	3.4	3.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.2	82.1	83.8	83.3	85.2	86.8	88.5	88.8	89.1	88.2	85.4	84.2
		MCWB	67.8	69.4	69.7	69.6	72.7	73.8	75.0	75.5	75.3	73.9	72.4	70.8
	2%	DB	80.7	80.5	81.2	81.4	83.1	84.5	86.5	86.8	86.7	85.9	83.4	81.4
		MCWB	68.5	69.2	69.6	70.0	71.6	72.6	74.4	74.6	74.4	73.7	72.0	69.8
	5%	DB	79.1	78.9	79.4	80.3	81.9	83.6	84.7	85.3	85.2	84.2	82.0	79.9
		MCWB	68.5	68.8	69.0	69.6	71.1	72.2	73.5	73.9	73.8	73.3	71.5	69.5
	10%	DB	77.4	77.2	78.1	79.0	81.0	82.2	83.6	84.1	84.0	82.8	80.9	78.8
		MCWB	68.1	68.1	68.6	69.1	70.7	71.5	72.9	73.2	73.2	72.6	71.1	69.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.2	73.8	73.5	74.2	74.9	75.5	77.2	77.5	78.0	77.2	75.6	73.7
		MCDB	76.7	77.1	78.8	78.5	80.8	82.9	83.8	84.7	84.5	82.5	80.7	79.1
	2%	WB	71.8	72.4	72.0	72.2	73.7	74.3	75.8	76.3	76.5	75.7	74.3	72.3
		MCDB	76.0	76.3	77.5	78.2	80.1	81.9	83.0	83.4	83.0	81.7	79.8	77.7
	5%	WB	70.8	71.2	71.0	71.2	72.8	73.2	74.9	75.4	75.4	75.0	73.3	71.4
		MCDB	75.5	75.5	76.5	77.5	79.4	81.0	82.1	82.3	82.0	81.2	79.0	76.9
	10%	WB	69.8	69.8	69.8	70.3	71.8	72.4	74.0	74.6	74.5	74.0	72.4	70.6
		MCDB	75.0	75.0	75.5	76.6	78.5	80.2	81.2	81.5	81.4	80.4	78.3	76.2
(35) Mean Daily Temperature Range	5% DB	MDBR	13.5	12.7	12.4	11.8	12.8	12.0	11.6	11.7	12.5	12.2	11.5	12.2
		MCDBR	15.4	14.9	14.7	13.4	14.2	13.4	12.7	12.7	13.5	13.6	13.1	14.0
	5% WB	MCWBR	7.9	7.9	7.5	6.7	6.8	6.1	5.7	5.6	6.2	6.3	6.5	7.4
		MCWBR	13.3	12.8	13.0	12.4	13.2	12.5	11.8	12.1	12.8	12.9	12.0	12.1
(40) Clear-Sky Solar Irradiance	taub		0.347	0.353	0.361	0.380	0.375	0.362	0.361	0.360	0.352	0.360	0.359	0.344
		taud	2.625	2.597	2.584	2.508	2.552	2.595	2.609	2.613	2.641	2.626	2.626	2.651
	Ebn at Noon		295	299	299	293	291	292	293	295	298	293	289	292
		Edh at Noon	28	30	32	35	33	31	31	31	30	29	28	26
(44) All-Sky Solar Radiation	RadAvg		1412	1591	1808	2050	2194	2284	2246	2168	2029	1711	1441	1300
	RadStd		61	104	177	119	118	62	51	85	60	47	35	101

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	+1.06	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+162	+164

Nomenclature: See separate page